APPROVED ASSEMBLIES

Membrane Type: Single Ply, Thermoplastic, PVC, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 gage Grade 33 Steel deck fastened to structural supports at a maximum span of 6 feet o.c. Steel deck shall be fastened with minimum ITW Buildex Teks 5 at a maximum spacing of 6 inches o.c. Side laps shall be fastened with ITW Buildex Teks 1 at a maximum spacing of 24 inches o.c.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(1): All layers of insulation simultaneously attached; membrane fully adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
Polyisocyanurate HP-H or InsulBase Minimum 2" thick	**1**	**1:1.6 ft²**

Note: All layers shall be simultaneously fastened; see top layer below for fasteners and density. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment. Single and multiple layers of insulation can be attached to base layer with Carlisle Syntec FAST 100-LV Adhesive.

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane: Sure-Flex, 50, 60 or 80 mil membrane fully adhered to the insulation using Low VOC PVC Bonding Adhesive applied to the substrate at a rate of 1 gal/60 ft.² The outside 1.5" of the roof cover lap is heat welded.

Maximum Design Pressure: -67.5 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, PVC, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 gage Grade 33 Steel deck fastened to structural supports at a maximum span of 6 feet o.c. Steel deck shall be fastened with minimum ITW Buildex Teks 5 at a maximum spacing of 6 inches o.c. Side laps shall be fastened with ITW Buildex Teks 1 at a maximum spacing of 24 inches o.c.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type C(2): All layers of insulation simultaneously attached; membrane fully adhered.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Base Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
ENRGY-3, AC Foam II, Polyisocyanurate HP-H, or InsulBase Minimum 1.5" thick	**N/A**	**N/A**

Top Insulation Layer	**Insulation Fasteners (Table 3)**	**Fastener Density/ft²**
Required over the insulations listed in Base Layer or optional over any of the insulations listed as Base or Top Layer:		
Dens Deck Prime Minimum ¼ " thick	**1, 2, 3**	**1:2 ft²**

Note: All layers shall be simultaneously fastened; see top layer below for fasteners and density. Insulation panels listed are minimum sizes and dimensions; if larger panels are used, the number of fasteners shall be increased maintaining the same fastener density. Please refer to Roofing Application Standard RAS 117 for insulation attachment. Single and multiple layers of insulation can be attached to base layer with Carlisle Syntec FAST 100-LV Adhesive.

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane: Sure-Flex Reinforced, 50, 60 or 80 mil membrane fully adhered to the insulation using Low VOC PVC Bonding Adhesive applied to the substrate at a rate of 1 gal/60 ft.² The outside 1.5" of the roof cover lap is heat welded.

Maximum Design Pressure: -45 psf (See General Limitation #7)

Membrane Type: Single Ply, Thermoplastic, PVC, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 gage Grade 33 Steel deck fastened to structural supports at a maximum span of 6 feet o.c. Steel deck shall be fastened with minimum ITW Buildex Teks 5 at a maximum spacing of 6 inches o.c. Side laps shall be fastened with ITW Buildex Teks 1 at a maximum spacing of 24 inches o.c.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(1): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
ENRGY-3, AC Foam II, Polyisocyanurate HP-H, or InsulBase Minimum 1.5" thick	**N/A**	**N/A**

Note: All layers of insulation and membrane shall be simultaneously attached. See membrane below for fasteners and density. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft. Single and multiple layers of insulation can be attached to the deck with FAST 100-LV Adhesive.

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane: Sure-Flex, Reinforced, secured through the preliminarily attached insulation as specified below.

Fastening #1: HP-X Fasteners with Piranha Plates 6" o.c. through the Sure-Flex Membrane in the lap in rows spaced 35" o.c. The outside 1.5" of the roof cover lap is heat welded.
Maximum Design Pressure -82.5 psf. (See General Limitation #7)

Fastening #2: HP-X Fasteners with Piranha Plates 12" o.c. through the Sure-Flex Membrane in the lap in rows spaced 35" o.c. The outside 1.5" of the roof cover lap is heat welded.
Maximum Design Pressure -52.5 psf. (See General Limitation #7)

Fastening #3: HP-Xtra Fasteners with Piranha Xtra Plates 12" o.c. through the Sure-Flex Membrane in the lap in rows spaced 35" o.c. The outside 1.5" of the roof cover lap is heat welded.
Maximum Design Pressure –60 psf. (See General Limitation #7)

Maximum Design Pressure: –See Fastening Options Above

Membrane Type: Single Ply, Thermoplastic, PVC, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 20 gage Grade 33 Steel deck fastened to structural supports at a maximum span of 6 feet o.c. Steel deck shall be fastened with minimum ITW Buildex Teks 5 at a maximum spacing of 6 inches o.c. Side laps shall be fastened with ITW Buildex Teks 1 at a maximum spacing of 24 inches o.c.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(2): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
ENRGY-3, AC Foam II, Polyisocyanurate HP-H, or InsulBase Minimum 1.5" thick	**N/A**	**N/A**

Note: All layers of insulation and membrane shall be simultaneously attached. See membrane below for fasteners and density. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft. Single and multiple layers of insulation can be attached to the deck with FAST 100-LV Adhesive.

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane: Sure-Flex, Reinforced, secured through the preliminarily attached insulation as specified below.

Fastening #1: HP-X Fasteners with Piranha Plates 6" o.c. through the Sure-Flex Membrane in the lap in rows spaced 75.5" o.c. The outside 1.5" of the roof cover lap is heat welded.
Maximum Design Pressure –52 psf. (See General Limitation #7)

Fastening #2: HP-Xtra Fasteners with Piranha Xtra Plates 12" o.c. through the Sure-Flex Membrane in the lap in rows spaced 75.5" o.c. The outside 1.5" of the roof cover lap is heat welded.
Maximum Design Pressure –45 psf. (See General Limitation #7)

Maximum Design Pressure: –See Fastening Options Above

Membrane Type: Single Ply, Thermoplastic, PVC, Reinforced

Deck Type 2I: Steel, Insulated

Deck Description: Minimum 22 gage ASTM A1008 or A653 SS Grade 80 Steel deck fastened to structural supports at a maximum span of 6 feet o.c. Steel deck shall be fastened with minimum ITW Buildex Teks 5 at a maximum spacing of 6 inches o.c. Side laps shall be fastened with ITW Buildex Teks 1 at a maximum spacing of 24 inches o.c.

This Tested Assembly has been analyzed for allowable deck stress. See Evidence Submittted Table.

System Type D(3): Membrane mechanically attached over preliminarily fastened insulation.

All General and System Limitations apply. Roof accessories not listed in Table 1 of this NOA are not approved and shall not be installed unless said accessories demonstrate compliance with prescriptive Florida Building Code requirements and are field fabricated utilizing the approved membranes listed in Table 1.

Insulation Layer	Insulation Fasteners (Table 3)	Fastener Density/ft^2
ENRGY-3, AC Foam II, Polyisocyanurate HP-H, or InsulBase, H-Shield, Insulfoam SP, R-Tech, R-Tech Fan-Fold, Securock		
Minimum 1.5" thick	**N/A**	**N/A**

Note: All layers of insulation and membrane shall be simultaneously attached. See membrane below for fasteners and density. Refer to Roofing Application Standard RAS 117 for insulation attachment requirements. Insulation shall have preliminary attachment, prior to the installation of the roofing membrane. At an application rate of two fasteners per board for insulation boards having no dimension greater than 4 ft., and four fasteners for any insulation board having no dimension greater than 8 ft. Single and multiple layers of insulation can be attached to the deck with FAST 100-LV Adhesive.

Vapor Retarder: (Optional) Any UL or FMRC approved vapor retarder applied to the roof deck or over a base layer of insulation.

Barrier: None.

Membrane: Sure-Flex, Reinforced, secured through the preliminarily attached insulation as specified below.

Fastening: HP-X Fasteners with Piranha Plates 12" o.c. through the Sure-Flex Membrane in the lap in rows spaced 75.5" o.c. The outside 1.5" of the roof cover lap is heat welded.

Maximum Design Pressure: -45 psf. (See General Limitation #7)